

Full-Stack & Generative AI Assessment Suite

A Technical Overview of Architecture, Supabase Implementation, Security Policies & Production Deployment

CANDIDATE	STACK	LIVE PRODUCTION URL	STATUS
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1. Executive Summary

This report outlines the end-to-end implementation of the full-stack engineering and Generative AI assessment suite consisting of three core technical projects. The suite demonstrates enterprise-grade serverless architecture, strict PostgreSQL security paradigms, real-time reactive data flows, and modern UX design.

Core Implementation Approach:

I engineered a modular full-stack architecture using React (JSX), Supabase PostgreSQL, and Node.js serverless functions, unified under a single evaluation portal.

For Task 1, I implemented Supabase Auth with strict Row-Level Security (RLS) data isolation and atomic PL/pgSQL stored procedures (RPC).

For Task 2, I developed an authentication flow with Google login that captures login telemetry into an audit table using live Supabase Realtime subscriptions.

For Task 3, I integrated Google Gemini (`gemini-3.6-flash`) through a secure serverless edge proxy to deliver a responsive, production-ready GenAI chatbot.

2. System Architecture & Project Organization

Each task is architected as an isolated, standalone application with its own frontend, backend runner, database migrations, and environment configurations, while also being dynamically unified into a production suite on Vercel:

Task	Domain & Functional Focus	Backend / Runtime	Database & AI
Task 1	Supabase Hands-On: Auth, RLS Policies, Database RPC & Edge Functions	Node.js Express & Supabase Edge Function	PostgreSQL RLS, 2 PL/pgSQL Stored Procedures
Task 2	Google OAuth, Telemetry Logger & Real-time Live Audit Trail	Node.js Edge Runner & Static App Host	<code>login_records</code> Table & Supabase Realtime Channels
Task 3	GenAI Chatbot: Multi-turn Context & Markdown Syntax Rendering	Vercel Serverless Function & Edge Handler	Google Gemini REST API (<code>gemini-3.6-flash</code>)
Unified Suite	Single Recruiter Evaluation Portal with Topbar Task Switcher	Vercel Edge Platform & Serverless API Routes	Unified Production Domain with SSL

3. Key Design Decisions

- Zero Mixed-Content Production Architecture:** Frontends on Vercel use HTTPS and automatically route backend calls to secure Serverless Edge API routes (`/api/*`), while falling back to port runners (`localhost:54321-54323`) during local development.
- Database-Enforced Access Control:** Security is enforced at the database engine level via PostgreSQL Row-Level Security (RLS), guaranteeing data safety independent of client-side code.
- API Key Isolation:** Generative AI tokens and Supabase Service Role keys remain strictly confidential on the serverless edge layer and are never compiled into client bundles.

4. Task 1: Supabase Hands-On Deep Dive

Task 1 demonstrates mastery of Supabase core infrastructure: Authentication, Row-Level Security (RLS), Database Remote Procedure Calls (RPC), and privileged Edge Functions.

A. Authentication & Session Management

- Implemented secure user registration and login using `@supabase/supabase-js`.
- Maintained persistent auth state through `supabase.auth.onAuthStateChange`, exposing active user ID (`auth.uid()`) and cryptographic JWT access tokens.
- Provided pre-configured test personas (Alice and Bob) to allow instant demonstration of cross-user authorization boundaries.

B. Database RPC (Stored Procedures)

To guarantee atomic calculations and reduce round-trip network latency, two PL/pgSQL functions were implemented in PostgreSQL and invoked via `supabase.rpc()`:

- `get_user_note_stats(target_user_id UUID)`: Aggregates total notes, private count, and last active timestamp for a user.
- `get_community_summary()`: Calculates platform-wide metrics (total public knowledge base count and active authors) atomically in a single query.

C. Row-Level Security (RLS) Policies

The `public.notes` table is secured using four discrete RLS policies:

```
-- 1. SELECT: Users view own notes OR public notes
CREATE POLICY "notes_select_policy"
ON public.notes FOR SELECT
USING (auth.uid() = user_id OR is_private = false);

-- 2. INSERT: Enforce authentic author assignment
CREATE POLICY "notes_insert_policy"
ON public.notes FOR INSERT
WITH CHECK (auth.uid() = user_id);

-- 3. UPDATE & DELETE: Owner-only modifications
CREATE POLICY "notes_update_policy"
ON public.notes FOR UPDATE USING (auth.uid() = user_id);

CREATE POLICY "notes_delete_policy"
ON public.notes FOR DELETE USING (auth.uid() = user_id);
```

Security Proof:

When Alice signs in and creates a private note, Bob is cryptographically blocked by PostgreSQL from querying or modifying Alice's record.

5. Task 2: Google Login & Real-Time Login Audit Trail

Task 2 fulfills the requirements for an OAuth-enabled authentication dashboard that captures detailed session telemetry and streams live login events to the user interface.

A. Telemetry Audit Table Schema

```
CREATE TABLE IF NOT EXISTS public.login_records (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID,
  email TEXT NOT NULL,
  user_name TEXT,
  avatar_url TEXT,
  user_agent TEXT,
  ip_address TEXT,
  logged_in_at TIMESTAMPTZ NOT NULL DEFAULT now()
);

CREATE INDEX IF NOT EXISTS idx_login_records_logged_in_at
ON public.login_records(logged_in_at DESC);
```

B. Dual-Path Recording & Real-Time WebSockets

- Dual-Path Resilience:** Every login triggers an Edge Function call (`/api/log-login`) to capture IP and user-agent metadata, paired with direct database fallback.
- Live Real-Time WebSockets:** Frontend subscribes to PostgreSQL changes via `supabase.channel().on('postgres_changes', ...)`. New sign-ins dynamically appear in under 150ms without page reloads.
- Device & Timestamp Parsing:** Converts user agents into human-readable browser tags (Chrome, Safari) with exact and relative timestamps ("Just now", "2m ago").

6. Task 3: Generative AI Chatbot (Google Gemini API)

Task 3 couples a responsive React chat interface with an Edge Function communicating securely with the Google Gemini API.

A. Edge Proxy Architecture & Secret Protection

Calling Google Gemini directly from React forces the `GEMINI_API_KEY` into client-side code, allowing anyone with browser DevTools to steal the key and drain API quotas.

The frontend communicates strictly with the serverless endpoint (`/api/chat-gemini`), which:

- Keeps `GEMINI_API_KEY` completely confidential on the serverless edge runtime.
- Enforces system instructions and persona constraints (`temperature: 0.7` , `maxOutputTokens: 2048`).
- Connects to the active `gemini-3.6-flash` model via Google Generative Language REST endpoints.

B. Conversational Interface Features

- Multi-Turn Context:** Automatically preserves conversation history so the model maintains contextual reasoning.
- Markdown & Code Highlighting:** Custom lightweight parser renders code blocks with monospace font, highlighted keywords, and a 1-click clipboard copy button.
- Adaptive Endpoint Routing:** Automatically switches between local development runner (port 54323) and Vercel serverless functions in production without manual code changes.

7. Production Deployment & Verification Matrix

All four builds, serverless handlers, and database policies have been verified and tested for production readiness:

Component	Verification Test	Result	Operational Notes
Unified Portal	<code>npm run build</code>	Passed (1.53s)	4,810 modules transformed, 0 warnings
Task 1 Standalone	<code>npm run build</code>	Passed (801ms)	Independent bundle verified
Task 2 Standalone	<code>npm run build</code>	Passed (802ms)	Independent bundle verified
Task 3 Standalone	<code>npm run build</code>	Passed (659ms)	Independent bundle verified
Node.js Backends	<code>node --check</code>	Passed (0 errors)	Syntax validated across all servers
Serverless API Routes	<code>/api/chat-gemini</code> , <code>/api/admin-action</code> , <code>/api/log-login</code>	Active & Live	Vercel Edge Serverless execution
Secret Protection	Git audit / <code>.gitignore</code>	100% Protected	All <code>.env</code> files excluded; zero secrets in Git

Summary for Evaluator:

The complete codebase is organized in clean isolated directories, protected against secret leaks, documented with step-by-step setup guides, and deployed as a unified live suite at <https://smaranai-tasks-mzk1.vercel.app>.